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Human-Centric Optimization of Workforce Scheduling under Fatigue

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Abstract Even though residential construction projects are delivered through labor-intensive tasks performed under tight deadlines, current project scheduling and staff planning models usually treat workforce capacity as fixed and overlook the operational consequences of human-centric factors. In this study, human-centric factors refer to labor fatigue, fairness in experiencing fatigue, fatigue-induced quality degradation, and fatigue-related absenteeism. To address this shortcoming, we introduce an integrated personnel staffing and project scheduling model for residential construction projects that gradually progresses to a human-centric-aware planning framework. By way of a simple yet innovative fatigue activation mechanism, we reveal the hidden operational impact of assigned workload on workforce fatigue experience. This mechanism allows us to examine how workload pressure influences subcontracting decisions, workload imbalance, quality, and workforce working capacity. We incorporate human-centric factors sequentially to isolate their impacts and to characterize how each mechanism reshapes staffing and scheduling decisions. Analytically, we show that the proposed extensions affect the model through distinct structural channels. Fairness incorporation restricts the feasible region by limiting excessive imbalance in fatigue exposure across the workforce; fatigue-induced quality degradation introduces endogenous rework costs; and absenteeism risk links workload allocation directly to the effective working capacity of the workforce through an endogenous fatigue–capacity feedback mechanism. In contrast to conventional scheduling models that treat worker capacity to perform allocated jobs as constant, our framework makes effective workforce capacity a function of fatigue experience. Contrary to what may be expected, incorporating sick-leave risk can improve the quality of work relative to a quality-only framework. This occurs because the model anticipates the future capacity loss associated with excessive fatigue exposure and, as a result, reallocates workload more conservatively, which in turn reduces fatigue experience across the workforce and generates a quality-enhancing spillover effect. Our findings highlight that, although current construction practice often treats human-centric factors as post-scheduling considerations, these factors operate as structural mechanisms that shape feasible, efficient, and reliable project delivery.

Key words: Fatigue; Absenteeism; Staffing; Scheduling; Fairness; Construction

1 Introduction

Construction projects are cost-intensive production systems where workforce planning plays a critical role in determining project success. In construction projects, labor costs account for about 30%-50% of total project costs (Larson and Gray 2014, McTague and Jergeas 2002), while shortages of skilled construction workers increas-

ingly constrain delivery capacity. Simultaneously, skilled laborers are the main productive resource through which construction schedules are executed and buildings are delivered. Hence, improving scheduling, staffing, and allocation of construction workers across project tasks is essential for achieving both timely project completion and operational efficiency. Not surprisingly, workforce planning has witnessed growing interest within operations management and construction literature. Common solutions include expanding training pipelines (Vairaktarakis and Winch 1999, Johari and Jha 2020), finding optimal staffing strategy (Gurvich et al. 2010, Ham et al. 2020), improving workforce mobility (Bard and Wan 2008, Zhu et al. 2016), and using novel scheduling models to coordinate workforce demand across tasks and locations (Milička et al. 2022). However, much of the current construction and operations management literature gives limited attention to how human-centric factors, such as fatigue and its consequences, impact project delivery. Therefore, in this article, we focus on human-centric factors as a novel operational factor impacting construction project performance. Specifically, we examine how worker fatigue, fatigue-related fairness, quality degradation, and risk of absenteeism affect staffing and scheduling decisions and project completion time.

The link between allocated workload and operational performance has drawn growing attention in the operations management literature, particularly in contexts where system performance depends on worker behavior. Classical personnel staffing and project scheduling models typically assume that staff productivity, task duration, and performance are exogenous and independent of assigned workload. However, a growing body of empirical studies reveals that performance in human-driven systems can be endogenously impacted by assigned workload (e.g., (Staats and Gino 2012, Tan and Netessine 2014, Bavafa and Jónasson 2024)) . Importantly, these workload-based impacts are often mediated by human-centric dynamics such as fatigue accumulation, cognitive strain, and behavioral reactions to continuous work-related pressure. These effects are particularly prominent in resource-constrained environments where timely completion of activities is critical. Prior studies document such effects across several settings, such as intensive care units, emergency departments, surgical operations, and field service teams (e.g., Shen et al. (2021), Kc and Terwiesch (2009), Shen et al. (2021), Batt and Terwiesch (2017), Bavafa and Jónasson (2024)). However, in construction settings, which is not considered an attractive sector for many workers due to tough working conditions (Kazaz and Ulubeyli 2007), typically workers are assisgned to several physically and cognitively demanding activities within a day and frequently operate under tight deadlines. Many construction tasks are labor-intensive and need continuous effort over long working hours, leading to a substantial physical and mental burden on construction workers. Such conditions can result in worker fatigue, which has been shown to impact quality of work, such as error frequency, service reliability, and post-service results (e.g., Kc and Terwiesch (2009), Needleman et al. (2011)).

Collectively, these articles indicate that workload influences operational performance not only directly but also indirectly through human-centric factors. However, despite these advances, limited focus has been on systematically incorporating such human-centric factors into construction staffing and scheduling frameworks. In this article, we study how human-centric factors influence worker performance and project staffing and scheduling decisions in construction settings. In this research, we conceptualize workload by modeling its influence on worker

fatigue instead of treating workload as a purely instantaneous measure at the system level. Particularly, fatigue is formulated as an endogenous intertemporal worker state that accumulates through workload and persists across time periods while gradually dissipating through recovery. Hence, this way of fairness representation distinguishes workload exposure from the resulting fatigue state, allowing labour with the same current workload to experience different fatigue levels based on their preceding workload and recovery histories. In most of the available literature, the workload of workers is typically evaluated at the system level instantaneously (e.g., Berry Jaeker and Tucker (2017), **?**), Kuntz et al. (2015)). In contrast, this research focuses on an individual-level dynamic fatigue state generated by current and historical workload exposure, which facilitates a more behaviorally relevant representation of the way that working pressure influences workers. Due to the nature of construction tasks and continuous workload pressures, incorporating individual-level accumulation impacts becomes particularly important.

The empirical evidence from the construction literature shows various factors contributing to the physical workload on workers. Among the various determinants of construction labour fatigue, we focus on workload pressure and recovery because both are directly shaped by operational planning decisions. This focus allows us to capture how staffing and scheduling decisions influence fatigue accumulation and recovery over time. Workload allocation in one time period raises the fatigue state of workers, while incomplete recovery causes part of this state to persist into subsequent periods. This dynamic mechanism provides operational levers to manage fatigue exposure through workload timing, recovery opportunities, workforce reallocation, and external sourcing. For example, although overall required workers may be difficult to modify in the short term, project managers often have more flexibility in redistributing workload allocation across workers to mitigate fatigue and enhance project efficiency. Thus, the formulation captures how current staffing decisions influence future labour condition and, through fatigue, affect quality performance and subsequent workforce availability. As fatigue exposure is shaped by allocated workload to workers, it also raises equity concerns regarding the distribution of fatigue burdens across workers. As personnel staffing decisions determine how cumulative fatigue develops among individual construction workers, the dynamic fatigue formulation also raises a distributive question of who bears the resulting fatigue pressure. Gilliland (1993) distinguishes two broad dimensions of fairness: distributive fairness and procedural fairness. Distributive fairness refers to the fairness of the outcome, including how rewards, resources, workloads, or burdens are allocated across individuals; see, for example, Geng et al. (2015) and Rea et al. (2021) for operations-management applications. Procedural fairness, by contrast, concerns the fairness of the decision making process. Operations management literature have highlighted the importance of procedural fairness for employee attitudes, employee attitudes, including organizational commitment (Wiesenfeld et al. 2007) and job satisfaction (Long et al. 2011). The fairness literature also identifies the third dimension, which focuses on the interpersonal side of organizational practices concerning the way that the organization behaves (interactional fairness) (Cohen-Charash and Spector 2001). In this article, fairness is formulated as distributive equity in cumulative fatigue pressures. As fatigue is an undesirable burden, the objective is not to make workers equally fatigued, instead the proposed model restricts the maximum cumulative fatigue pressure experienced by a worker relative to the nominal availability of that worker.

1.1 Motivation from Discussions with Construction Practitioners

Discussions with construction site managers and project managers in Australia motivated the authors regarding the practical challenges of worker shortage, delivery pressures, workforce fatigue, and their absenteeism in construction projects. During these discussions, it was revealed that residential construction projects, including those delivered by larger companies, continue to rely heavily on project schedules as the central tool for coordinating work. However, human-factor considerations (e.g., fatigue of worker, the distribution of fatigue burden across workers, quality degradation, and absenteeism risk) are rarely embedded in these planning tools in a systematic way. Instead, they mentioned that these factors are often managed by reactive adjustments within construction sites once disruptions or absenteeism have already emerged.

The discussions also indicated that fairness in construction projects is commonly approached from a compliance perspective. For example, relying on employment regulations and workplace requirements, such as overtime payments, limits on maximum fortnightly workload, and other conditions governed by Fair Work arrangements. Although these mechanisms are essential to protect minimum employment standards, they do not provide a project-level proactive planning framework to manage how the cumulative fatigue pressure is distributed across workforce. Hence, a construction schedule may respect regulatory requirements while still concentrating fatigue burden on a subset of workers, potentially raising subsequent absenteeism risk and quality-related consequences.

Similarly, construction practitioners noted that fatigue of workers and their requests for sick leave are not usually incorporated into their scheduling processes. Commonly, whenever absenteeism occurs, the site manager or project manager responds by relying on their experience by reallocating workers, if the deadline is tight, increasing overtime, engaging subcontractors more, or revising short-term site plans. These solutions may help maintain project progress, but they are mainly reactive and can pose additional costs, quality, safety, and workforce well-being implications. This highlights a critical operational gap in current construction environments: construction schedules are designed to control project timely delivery and resource feasibility, but they often do not anticipate how decisions regarding personnel staffing and project scheduling contribute to the accumulation and recovery of labour fatigue and how the resulting fatigue state subsequently influence distributive fairness, quality degradation, and future worker availability.

During the discussions, practitioners highlighted that wearable technologies could provide useful information for identifying fatigue or injury risk in construction projects. When they were asked whether the information regarding the physical condition of workers could influence scheduling decisions, they noted that such information could lead to decisions, such as giving rest to fatigued workers, revising on-site workload allocation, or, in some cases, where several key workers are involved, adjusting the project schedule. However, the discussions revealed that these decisions are made through a sequential process order: the project schedule is first developed, and fatigue monitoring information is then monitored during execution using wearable technologies or other monitoring systems. This indicates that such technologies can be highly valuable to detect fatigue risk among workers, but their role is often activated once the project schedule has already been made. This observation motivated the need for developing an optimization framework in which fatigue is not only monitored during execution but is formulated

proactively as an evolving worker state. In this framework, current workload contributes to fatigue accumulation, recovery reduces the residual state over time, and the resulting fatigue affects quality, fairness, and subsequent staff availability.

1.2 Research Questions and Contributions

Collectively, despite the growth of attention to human-centric dynamics, their integration into construction staffing and scheduling decisions remains limited. In particular, there is a need to understand how labour conditions evolve as a consequence of operational decisions and how the resulting influence fairness, quality degradation, and worker availability feed back into project planning. In light of these gaps, this paper aims to answer the following questions.

- What modeling architecture enables workload burden, dynamic fatigue accumulation and recovery, and the resulting human-centric consequences such as fairness, quality degradation, and absenteeism to be incorporated into integrated staffing and scheduling decisions without compromising tractability?
- What are the explicit and system-level effects of incorporating human-centric factors on personnel staffing decisions, project scheduling decisions, and project performance outcomes?

This article provides three main contributions to address the research questions. First, we introduce a new optimization framework in which fatigue is incorporated as an endogenous intertemporal worker state produced by workload decisions. The proposed formulation distinguishes between workload burden and the resulting fatigue state. Moreover, accumulated fatigue in one period can be persist into subsequent periods while gradually dissipating through recovery. Hence, labours with identical workloads may experience different levels of fatigue based on their preceding workload and recovery histories. Moderate and high fatigue categories are incorporated as classifications of the continuous fatigue state. This model establishes a direct link between current personnel staffing decisions and future worker condition while maintaining a mixed-integer linear structure. It also reveals how workload assignment and recovery opportunities can affect subsequent workforce decisions, such the allocation of internal workers and the use of subcontractors.

Second, we represent the influence of the endogenous fatigue state into the optimization framework through a stepwise modeling approach. This approach allows to capture and analyze the impact of human-centric performance factors, such as quality degradation, absenteeism, and fairness, independently and jointly on personnel staffing and scheduling decisions.

We analyze how each factor affects operational decisions, which uncovers nontrivial interactions between human-centric factors. For example, in contrast to expectations, incorporating risk of absenteeism into the baseline model resulted in lower quality loss than inserting the quality loss structure into the baseline model. We also find that when considering human-centric factors, focusing only on fewer quality loss or fair workload assignments can lead to higher project completion time compared to when sick-leave risk is incorporated. We also demonstrate that human-centric dynamics are not monotonic: incorporating additional human factors does not always lead to uniformly higher or lower performance, but instead reshapes the existing trade-offs among project delivery time, quality, and staff sustainability in subtle and occasionally counterintuitive ways.

Third, we develop a set of structural findings that characterize how the dynamic fatigue structure affects staff planning decisions. We first demonstrate how an additional unit of workload propagates through future fatigue states and derive the minimum recovery time needed for a worker to leave an elevated fatigue regime. Then, we characterize fatigue smoothing over a temporally flexible work window and find that the relevant operational principle is to smooth the worker's fatigue trajectory instead of necessarily equalize working hours. A complementary finding demonstrates that earlier workload carries a higher cumulative within-horizon fatigue pressure because its impacts persist over more subsequent periods. For the proposed integrated model, we establish that higher fatigue persistence weakly contracts the feasible set of personnel staffing and project scheduling decisions and as a results cannot improve the optimal project completion. Finally, we derive a sufficient condition under which replacing fatigued internal work with subcontractors is noninferior even before nominal internal capacity is exhausted. Collectively, these findings identify the mechanisms that assigned workload, recovery opportunity, and recruitment decisions become endogenous responses to worker fatigue. Supplementary findings further characterize the fairness tolerance and rework-cost thresholds at which the associated human-centric mechanisms begin to exclude optimal solution. Together, these results contribute to human-centric-aware operations management in construction projects by providing an integrated analytical framework connecting operational decisions, evolving worker condition, and project performance in construction staffing and scheduling.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the model development. Section 4 reports the experimental results. Finally, Section 5 concludes the paper and outlines directions for future research. All proofs are provided in the Online Appendix.

2 Literature Review

This paper contributes to three streams of literature: integrated personnel staffing and project scheduling, fatigue in construction and operations, and the operational consequences of fatigue and other human-centric factors.

2.1 Integrated Personnel Staffing and Project Scheduling

The literature on integrated staffing and scheduling has evolved from sequential project planning approaches toward models that jointly determine task schedules, workforce allocation, and workforce composition. Early studies, such as Heimerl and Kolisch (2010), demonstrate the advantages of centralized resource management and show that integrated approaches can reduce project costs relative to decentralized planning. Subsequent work jointly optimizes project schedules and staff rosters while incorporating regular, overtime, and temporary employment strategies (Maenhout and Vanhoucke 2016), and more recent studies develop advanced solution methods for integrated personnel staffing and scheduling problems (Hu et al. 2025).

Recent work in this stream has also begun to incorporate aspects of human performance. For example, Li et al. (2023) incorporate learning effects and examine their implications for task duration, makespan, and productivity. However, broader human-centric dimensions, including fatigue, sick leave, and fairness, remain largely unexplored both in integrated staffing and scheduling models and, more broadly, in the project scheduling literature. Thus, while the existing literature demonstrates the efficiency benefits of integrating staffing and scheduling decisions, it

provides limited guidance on how evolving worker conditions can be incorporated into such decisions. Our work contributes to this stream by embedding fatigue and its operational consequences directly within an integrated staffing and scheduling framework.

2.2 Fatigue in Construction and Operations

Construction workers are systematically exposed to occupational fatigue due to the physically demanding nature of their tasks, which are often performed in ergonomically demanding postures and over extended working hours (Aryal et al. 2017). Such conditions are associated with increased error rates, workplace accidents, and adverse occupational health outcomes (Wang et al. 2023). Accordingly, a growing literature examines the measurement, drivers, and mitigation of fatigue among construction workers (Zong et al. 2024). Aryal et al. (2017) use physiological indicators such as heart rate and skin temperature to measure fatigue in laboratory-based construction tasks, while Lee et al. (2017) provide field-based evidence from roofing activities and document changes in physiological measures as fatigue accumulates over consecutive workdays. Zhang et al. (2023) further combine physiological indicators with behavioral performance measures and show that physical and mental fatigue impair cognitive and motor performance and increase unsafe behavior. Other studies investigate the underlying drivers of fatigue, which include work-related, personal, and environmental factors (Zong et al. 2024). Among these, task characteristics, working hours, and workload intensity are particularly relevant to operational planning (Chan et al. 2012). A related stream examines wearable sensing technologies for real-time fatigue monitoring (Aryal et al. 2017, Ahn et al. 2019).

From an operations management perspective, workload and fatigue are increasingly recognized as important determinants of individual and system performance. Workers may respond to high system load by accelerating service or modifying task execution, but such responses can reduce service quality and worsen downstream outcomes (Batt and Terwiesch 2017, Kc and Terwiesch 2009). Workload effects can also accumulate over time and differ across individuals. For example, higher daily workload can lead to longer service times and poorer outcomes (Shen et al. 2021), while fatigue can increase both average service times and performance variability (Bavafa and Jónasson 2024). Despite these advances, much of this literature focuses on service settings or on empirically measuring fatigue and its consequences. It provides limited guidance on how fatigue can be proactively incorporated into project-based staffing and scheduling decisions. Our study addresses this gap by modeling fatigue as an endogenous worker state that evolves with workload and recovery and directly influences operational decisions.

2.3 Operational Consequences of Fatigue and Human-Centric Considerations

The consequences of heavy workload have been studied across healthcare (Farid et al. 2020, Rusnock et al. 2017, Schoenfelder et al. 2020, Piri et al. 2022, Li and Piri 2025), manufacturing (Moon et al. 2023, Leva et al. 2022), and construction (Tonnon et al. 2019, Robroek et al. 2017). Recent empirical evidence further shows that cumulative workload and overtime exposure can increase burnout and workforce attrition (Liu et al. 2026). Because the consequences of workload are broad and context dependent, we focus on three operationally relevant dimensions: quality degradation, absenteeism, and fairness.

The first is quality degradation. Empirical studies consistently show that elevated workload can undermine service quality and operational reliability. For example, Gurses et al. (2009) show in healthcare settings that high workload reduces service quality and safety. In construction, physical and mental fatigue can impair task execution by affecting workers' cognitive and motor abilities, increasing risk-taking behavior, and raising the likelihood of operational errors (Zhang et al. 2023). These findings provide empirical support for linking worker fatigue to quality degradation in operational planning.

The second is absenteeism. Unruh et al. (2007) examine absenteeism under varying workload conditions and show that its interaction with high workload can lead to substantial deterioration in outcomes. Complementing this evidence, Sagherian et al. (2019) find that fatigue significantly increases the likelihood of long-term sick leave, with fatigued workers exhibiting approximately 35% higher risk. In the construction sector, **?** further show that health-related conditions contribute to productivity losses beyond the direct effects of absenteeism. These findings motivate our modeling of fatigue as a driver of future effective workforce capacity through increased sick-leave risk.

The third dimension is fairness in workload allocation. Existing studies examine fairness from several perspectives in resource and task allocation. Cai et al. (2024) develop a target-based approach that prioritizes the least satisfied individual, while Fu and Liu (2021) examine fairness-aware task allocation and the trade-off between cost efficiency and workload distribution. Similarly, Mahmoodian et al. (2025) show that improving equity among heterogeneous drivers requires controlled deviations from the least-cost solution. These studies establish fairness as an important operational consideration, but fairness is rarely incorporated jointly with fatigue, quality degradation, and absenteeism in project staffing and scheduling models.

Taken together, the literature provides substantial evidence that staffing and scheduling integration can improve operational efficiency and that workload and fatigue can affect worker performance, quality, absenteeism, and workload equity. However, these streams remain largely separate. Integrated staffing and scheduling models primarily emphasize efficiency, while human-centric factors are generally studied in isolation or as outcomes rather than as endogenous components of operational planning. We bridge these streams by developing an integrated framework in which staffing and scheduling decisions determine fatigue accumulation and recovery, and the resulting fatigue state influences fairness, quality degradation, and future workforce capacity.

3 Mathematical model

In this paper, we develop an integrated personnel staffing and project scheduling model that explicitly accounts for the consequences of labor fatigue. The mathematical model is constructed sequentially. We first formulate the baseline model for the integration of personnel staffing and project scheduling. Then, an endogenous fatigue state is introduced that accumulates through workload pressure and dissipates through recovery. The resulting fatigue state subsequently influences three distinct human-centric outcomes: fairness in cumulative fatigue burden, quality degradation, and absenteeism risk. A central feature of this formulation is the separation between workload exposure, the resulting fatigue state, and its consequences:

$$\text{workload exposure} \longrightarrow \text{fatigue state} \longrightarrow \text{operational consequences}$$

This sequential structure simulate a structure tightened to reality of construction project in a way that staffing and scheduling decisions not only affect current resource utilization but also subsequent worker condition and project performance. Figure 1 demonstrates a conceptual overview of the problem we have considered in this paper. Traditional construction planning typically integrates project activities, workforce capacity, and project scheduling decisions with mainly emphasizing on project-level outcomes such as project completion time and cost. In contrast, our proposed human-centric framework explicitly recognizes that personnel staffing and project scheduling decisions can also affect labor conditions and, as a result, project performance. On the other hand, our proposed framework jointly considers activity scheduling and workforce allocation while incorporating skill of workers and their availability, as demonstrated in Figure 1. The resulting workload allocation affect worker fatigue, which is subsequently linked to quality shortfall, fairness in the distribution of fatigue pressure, and absenteeism risk. Therefore, these human-centric consequences are incorporated within the planning problem rather than being treated only as outcomes observed after the schedule has been developed. The following subsections develop this integrated formulation sequentially, beginning with the baseline staffing and scheduling model.

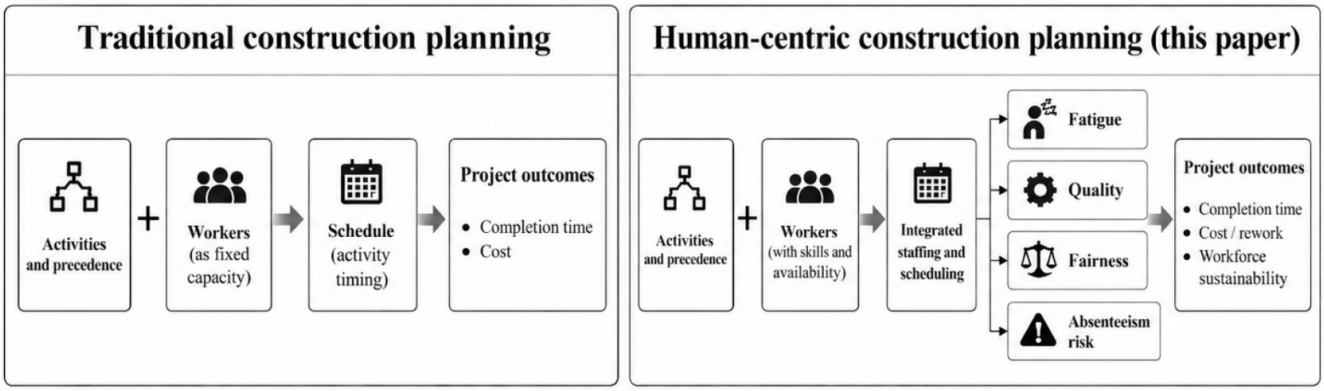


Figure 1 Conceptual comparison of traditional and human-centric construction planning.

3.1 Baseline project scheduling and workforce allocation

Let $\mathscr{I}$ denote the set of project activities, and let $I \in \mathscr{I}$ denote a dummy completion activity with zero duration and zero resource requirement. The start time of activity I therefore represents the project completion time. Let $\mathscr{D} = \{1, \ldots, D\}$ denote the planning horizon in days, $\mathscr{W}$ the set of internal workers, and $\mathscr{S}$ the set of skills. For each activity $i \in \mathscr{I}$, let p_i denote its duration and $\mathscr{P}(i)$ its set of immediate predecessors. Let n_{is} denote the amount of skill-s labor required on each day that activity i is in progress. For each worker $w \in \mathscr{W}$ and day $d \in \mathscr{D}$, let h_{wd} denote the worker's available working capacity, and let q_{ws} be a binary parameter equal to one if worker w possesses skill s and zero otherwise. Let c_s^{int} and c_s^{sub} denote the unit costs of internal and subcontracted labor with skill s, respectively. Finally, let C^{del} denote the unit cost of project delay, due the promised project due date, and b the available project budget.

The binary decision variable X_{id} equals one if activity i starts on day d, and A_{id} equals one if activity i is in progress on day d. The continuous decision variable Y_{wisd} denotes the amount of labor provided by worker w to

activity i using skill s on day d, while O_{isd} denotes the amount of subcontracted labor with skill s assigned to activity i on day d. The variable U_{wd} denotes the total workload assigned to worker w on day d, and T denotes the project delay beyond the promised due date. The baseline model is formulated as follows.

$$\min \quad Z = \sum_{d\in\mathscr{D}} dX_{Id} \tag{1}$$

$$\sum_{d\in\mathscr{D}} X_{id} = 1 \qquad \forall i \in \mathscr{I} \tag{2}$$

$$\sum_{d\in\mathscr{D}} dX_{id} + p_i \le \sum_{d\in\mathscr{D}} dX_{jd} \qquad \forall j \in \mathscr{I},\ \forall i \in \mathscr{P}\mathscr{r}(j) \tag{3}$$

$$A_{id} \le \sum_{\tau=\max\{1,d-p_i+1\}}^{d} X_{i\tau} \qquad \forall i \in \mathscr{I} : p_i > 0,\ d \in \mathscr{D} \tag{4}$$

$$\sum_{\tau=d}^{\min\{D,d+p_i-1\}} A_{i\tau} \ge p_i X_{id} \qquad \forall i \in \mathscr{I} : p_i > 0,\ d \in \mathscr{D} \tag{5}$$

$$\sum_{w\in\mathscr{W}} Y_{wisd} + O_{isd} = n_{is} A_{id} \qquad \forall i \in \mathscr{I},\ s \in \mathscr{S},\ d \in \mathscr{D} \tag{6}$$

$$\sum_{d\in\mathscr{D}} \sum_{i\in\mathscr{I}} \sum_{s\in\mathscr{S}} \left(c_s^{\text{int}} \sum_{w\in\mathscr{W}} Y_{wisd} + c_s^{\text{sub}} O_{isd} \right) + C^{\text{del}} T \le b \tag{7}$$

$$T \ge \sum_{d\in\mathscr{D}} dX_{Id} - \text{due} \tag{8}$$

$$U_{wd} = \sum_{i\in\mathscr{I}} \sum_{s\in\mathscr{S}} Y_{wisd} \qquad \forall w \in \mathscr{W},\ d \in \mathscr{D} \tag{9}$$

$$Y_{wisd} \le h_{wd} q_{ws} A_{id} \qquad \forall w \in \mathscr{W},\ i \in \mathscr{I},\ s \in \mathscr{S},\ d \in \mathscr{D} \tag{10}$$

$$U_{wd} \le h_{wd} \qquad \forall w \in \mathscr{W},\ d \in \mathscr{D} \tag{11}$$

$$X_{id}, A_{id} \in \{0,1\} \qquad \forall i \in \mathscr{I}, d \in \mathscr{D} \tag{12}$$

$$Y_{wisd} \ge 0 \qquad \forall w \in \mathscr{S}, i \in \mathscr{I}, s \in \mathscr{S}, d \in \mathscr{D} \tag{13}$$

$$O_{isd} \ge 0 \qquad \forall i \in \mathscr{I}, s \in \mathscr{S}, d \in \mathscr{D} \tag{14}$$

$$U_{wd}, T \ge 0 \qquad \forall w \in \mathscr{W}, d \in \mathscr{D} \tag{15}$$

Objective function (1) minimizes project completion time. Constraint (2) ensures that each task has a unique start period, while Constraint (3) enforces technological precedence relations. Constraints (4) and (5) link start decisions with task execution and ensure non-preemptive processing of tasks. Constraint (6) ensures that the skill requirements of each in-progress activity are satisfied through internal workers, subcontracted labor, or a combination of the two. Constraint (7) ensures that internal labor, subcontracting, and delay penalty costs remain within the project budget, while Constraint (8) defines project delay relative to the promised due date. Constraint (9) aggregates the daily assignments of each worker. Constraint (10) restricts internal workload allocation to workers who possess the required skill and to periods during which the corresponding activity is in progress. Finally, Constraint (11) restricts assigned workload to the daily availability of each worker, and Constraints (12) to (15) define the variable domains.

3.2 Dynamic fatigue accumulation and recovery

The baseline model assumes that the capacity of workers on a given day is unaffected by the workload experienced on previous days. We relax this assumption by introducing fatigue as a latent state that contains the impacts of the previous workload forward through time.

Let $E_{wd} \geq 0$ denote the workload exposure contributing to the fatigue of worker w on day d. We define

$$E_{wd} = \kappa U_{wd} \qquad \forall w \in \mathscr{W},\ d \in \mathscr{D} \tag{16}$$

where $\kappa > 0$ converts allocated workload into units of fatigue exposure. Let $F_{wd} \geq 0$ denote the effective fatigue state of worker w on day d by considering the residual fatigue burden carried from the previous period and the new workload pressure generated on day d. Let $F_w^0 \geq 0$ demonstrates the fatigue state immediately before the planning horizon. Fatigue evolves according to the equation below.

$$F_{wd} = \rho F_{w,d-1} + E_{wd} \qquad \forall w \in \mathscr{W},\ d \in \mathscr{D} \tag{17}$$

where $F_{w0} = F_w^0$ and $0 \leq \rho < 1$ denotes the fraction of previous fatigue that remains after inter-period recovery. If worker w receives no workload on day d, Equation (17) reduces to $F_{wd} = \rho F_{w,d-1}$, so fatigue declines through recovery. In contrast, repeated workload assignments adds new fatigue exposure before previously accumulated fatigue has fully dissipated. Therefore, the impact of consecutive working days on workers is captured naturally through cumulative fatigue over time. Repeated substitution yields

$$F_{wd} = \rho^d F_w^0 + \kappa \sum_{\tau=1}^{d} \rho^{d-\tau} U_{w\tau} \tag{18}$$

Equation (18) explicitly demonstrate the intertemporal nature of fatigue exposure. The fatigue state on any day is defined not only based on the assigned workload on that day but also on the entire preceding workload trajectory. The contribution of earlier workload pressure diminishes geometrically through recovery. The following result quantifies this propagation mechanism.

LEMMA 1. *For any worker w, the fatigue state on day d can be shown as*

$$F_{wd} = \rho^d F_w^0 + \kappa \sum_{\tau=1}^{d} \rho^{d-\tau} U_{w\tau}.$$

Consequently, increasing workload on day d by $\delta > 0$, while keeping all other workload assignments fixed, increases the fatigue state of that worker on day $d + \ell$ by

$$\Delta F_{w,d+\ell} = \kappa \rho^{\ell} \delta \qquad \ell = 0, 1, \ldots, D - d$$

Lemma 1 demonstrates that an additional unit of allocated workload increases current fatigue, with part of this impact carrying over to subsequent days. Its largest impact occurs in the period in which the work is performed, while the residual impact persists into subsequent days at rate of ρ. Therefore, the same current allocated workload can produce different fatigue states on workers according to the preceding workload and recovery history of each worker.

As the scale of a latent state is arbitrary, κ may be normalized to one without loss of generality when an independent empirical fatigue scale is not available. If fatigue measurements allow direct calibration, κ and the fatigue thresholds introduced below can instead be estimated jointly. For tractable links between the continuous fatigue state and its operational results, we distinguish low, moderate, and high fatigue regimes. Let $\bar{F}^{\text{mod}} < \bar{F}^{\text{high}}$ denote the corresponding fatigue state thresholds. Note that, fatigue is classified as low when $F_{wd} < \bar{F}^{\text{mod}}$. The continuous fatigue state also generates a direct characterization of recovery. Let $\bar{F} > 0$ be any fatigue-state threshold. If a labor receives no workload over successive periods, the fatigue state evolves only through the recovery component of Equation (17). This yeilds the following result.

PROPOSITION 1 **(Minimum recovery requirement)**. *Suppose* $0 < \rho < 1$, $F_{wt} > \bar{F} > 0$, *and worker w receives no workload during the following r consecutive periods. Then* $F_{w,t+r} = \rho^r F_{wt}$. *The minimum number of consecutive recovery periods needed for the fatigue state to fall to or below the threshold* $\bar{F}$ *is*

$$r^*(F_{wt}, \bar{F}) = \left\lceil \frac{\ln(\bar{F}/F_{wt})}{\ln \rho} \right\rceil.$$

Proposition 1 replaces an exogenous consecutive working day rule with an endogenous recovery period requirement. The amount of rest needed to leave a fatigue state depends jointly on the current fatigue state of worker and on the persistence parameter ρ. For example, setting $\bar{F} = \bar{F}^{\text{high}}$ provides the number of recovery periods required to leave the high fatigue state, whereas $\bar{F} = \bar{F}^{\text{mod}}$ provides the corresponding requirement to leave moderate fatigue state. The binary decision variables α_{wd} and β_{wd} denote whether worker w has reached at least the moderate and high fatigue states, respectively.

A valid day-specific upper bound on the fatigue state is evaluated recursively from

$$\begin{aligned}
&\overline{F}_{w0} = F_w^0 \\
&\overline{F}_{wd} = \rho \overline{F}_{w,d-1} + \kappa h_{wd} && \forall w \in \mathscr{W},\ d \in \mathscr{D} \qquad (19)
\end{aligned}$$

Let define $M_{wd}^{\text{mod}} = \max\{0, \overline{F}_{wd} - \bar{F}^{\text{mod}}\}$, and $M_{wd}^{\text{high}} = \max\{0, \overline{F}_{wd} - \bar{F}^{\text{high}}\}$. The fatigue states are classified through following constraints.

$$F_{wd} \le \bar{F}^{\text{mod}} + M_{wd}^{\text{mod}} \alpha_{wd} \qquad \forall w \in \mathscr{W}, d \in \mathscr{D} \qquad (20)$$

$$F_{wd} \ge \bar{F}^{\text{mod}} \alpha_{wd} \qquad \forall w \in \mathscr{W}, d \in \mathscr{D} \qquad (21)$$

$$F_{wd} \le \bar{F}^{\text{high}} + M_{wd}^{\text{high}} \beta_{wd} \qquad \forall w \in \mathscr{W}, d \in \mathscr{D} \qquad (22)$$

$$F_{wd} \ge \bar{F}^{\text{high}} \beta_{wd} \qquad \forall w \in \mathscr{W}, d \in \mathscr{D} \qquad (23)$$

$$\beta_{wd} \le \alpha_{wd} \qquad \forall w \in \mathscr{W}, d \in \mathscr{D} \qquad (24)$$

$$\alpha_{wd}, \beta_{wd} \in \{0, 1\} \qquad \forall w \in \mathscr{W}, d \in \mathscr{D} \qquad (25)$$

At the exact threshold values, the lower fatigue state is adopted as the tie convention. Aside from this boundary convention, $\alpha_{wd} = 0$ indicates low fatigue, $\alpha_{wd} = 1, \beta_{wd} = 0$ indicates moderate fatigue, and $\beta_{wd} = 1$ indicates high

fatigue. It is important to note that these binary variables do not define fatigue. Fatigue remains the continuous state F_{wd}; the regime indicators provide a tractable piecewise representation of how various consequences respond to that that particular fatigue state.

Beyond describing fatigue accumulation and recovery, the state dynamics also imply a scheduling principle for temporally flexible workloads. Consider a worker who must perform a fixed amount of work over a sequence of time periods, but for whom the allocation of that work across the periods can be changed without influencing project precedence logic or completion time. Then, the relevant smoothing objective is not necessarily to equalize assigned workloads, but to control the maximum fatigue state generated by those workloads.

PROPOSITION 2 **(Fatigue smoothing over a flexible work window)**. *Consider n consecutive time periods over which a fixed workload Q must be allocated to one worker. Suppose that workload can be freely reallocated in the time window without influencing project feasibility and that daily capacity limits are nonbinding. Let f denote the fatigue state immediately before the window. The minimum achievable peak fatigue state is*

$$F^{\mathrm{pk},*} = \max\left\{\rho f,\ \frac{\kappa Q + \rho f}{n-(n-1)\rho}\right\}.$$

If

$$\frac{\kappa Q + \rho f}{n-(n-1)\rho} > \rho f,$$

the optimal allocation equalizes the fatigue state across the work window:

$$F_1^* = \cdots = F_n^* = F^{\mathrm{pk},*},$$

and is produced by

$$U_1^* = \frac{F^{\mathrm{pk},*} - \rho f}{\kappa}, \qquad U_t^* = \frac{(1-\rho)F^{\mathrm{pk},*}}{\kappa}, \quad t = 2,\ldots,n$$

Proposition 2 provides a critical distinction between workload smoothing and fatigue smoothing. Equal working hours are not generally needed. Instead, when workload timing is sufficiently flexible, the minimum-peak solution equalizes the fatigue state of workers. If the worker enters the window relatively well recovered, the first period may accommodate more workload as part of the resulting fatigue dissipates before subsequent periods. If the worker enters with substantial residual fatigue, early workload intensity decreases, while the benefits of recovery become more valuable. This proposition explains a structural benchmark instead of asserting that every project schedule must exhibit this pattern. In the integrated mathematical model, precedence relations, skill requirements, daily capacity, and budget cap may constraints the ability to achieve the fatigue equalizing allocation.

COROLLARY 1 **(Fatigue-safe workload over a flexible window)**. *Under the conditions of Proposition 2, there exists a workload allocation satisfying*

$$F_t \le \bar{F} \qquad t = 1,\ldots,n$$

if and only if

$$\rho f \le \bar{F}$$

and

$$Q \le \frac{[n-(n-1)\rho]\bar{F} - \rho f}{\kappa}$$

Corollary 1 provides a dynamic analogue of labor capacity. Unlike nominal daily capacity, the amount of workload that can be done without exceeding a fatigue threshold depends on the initial state of worker, the length of the available work window, and the recovery rate.

Collectively, Lemma 1, Proposition 1, and Proposition 2 establish the three mechanisms that distinguish the dynamic formulation from static workload thresholds. Meaning that workload has persistent consequences, rest has an endogenous recovery value, and workload flexibility can be used strategically to control fatigue exposure.

3.3 Fairness in cumulative fatigue burden

In this article, we represent fairness as distributive equity in the cumulative fatigue pressure generated by staffing decisions. As fatigue is an undesirable outcome, fairness is not interpreted as making workers equally fatigued. Instead, the mathematical model follows a worst-off protection principle by restricting the maximum fatigue pressure borne by any labor relative to that worker's nominal availability.

For worker w, let define cumulative fatigue pressure as

$$B_w^F = \sum_{d \in \mathscr{D}} F_{wd} \qquad \forall w \in \mathscr{W} \tag{26}$$

This measure captures both intensity and persistence of fatigue. Two labors may reach the same peak fatigue level but experience different cumulative fatigue pressures if one remains fatigued for a longer time. The cumulative fatigue pressure can also be stated directly in terms of the timing of the worker's workload. This representation clarifies the reason for when two project schedules with the same total assigned workloads may generate different cumulative fatigue pressures.

PROPOSITION 3 **(Temporal weighting of workload)**. *For $0 < \rho < 1$, worker w's cumulative fatigue pressure satisfies*

$$B_w^F = \frac{\rho(1-\rho^D)}{1-\rho} F_w^0 + \kappa \sum_{d=1}^{D} \omega_d U_{wd},$$

where

$$\omega_d = \frac{1-\rho^{D-d+1}}{1-\rho} = \sum_{\ell=0}^{D-d} \rho^\ell$$

Furthermore,

$$\omega_d - \omega_{d+1} = \rho^{D-d} > 0, \qquad d < D.$$

Therefore, holding all other decisions fixed, one unit of workload performed earlier contributes more to the cumulative fatigue pressure than the same unit performed later.

Proposition 3 demonstrates that the dynamic fatigue model endogenously assigns a temporal weight to internal workload. Earlier workload stays in the fatigue state for more time and as a result contributes more to the cumulative fatigue burden used in the fairness mechanism. If $\rho = 0$, all temporal weights collapse to one and this intertemporal impact disappears. The result should be interpreted as a horizon-dependent accumulation effect rather than as a claim that late project workload is intrinsically less harmful.

Labors can differ in their exogenous availability over the planning horizon. Therefore, we define total nominal capacity as

$$C_w = \sum_{d \in \mathscr{D}} h_{wd} \qquad \forall w \in \mathscr{W} \tag{27}$$

For $C_w > 0$, the capacity-normalized fatigue pressure is

$$\eta_w = \frac{B_w^F}{C_w}.$$

Fairness is imposed through

$$B_w^F \leq f^{\max} C_w \qquad \forall w \in \mathscr{W} : C_w > 0 \tag{28}$$

where $f^{\max}$ indicates the maximum allowable normalized cumulative fatigue pressure. The use of nominal capacity in (27) is deliberate. Normalizing by fatigue-based capacity would make the fairness denominator itself endogenous to fatigue and generate a circular equity measure. Constraint (28) has a Rawlsian or minimax interpretation for an undesirable fatigue burden: it preserves the worker with the largest normalized fatigue burden. Unlike a range-based equity measure, it cannot improve the fairness by raising the fatigue of a lightly burdened worker.

Historical workload enters to the model naturally through F_w^0. A labor who begins the construction project with residual fatigue therefore has less room for additional fatigue accumulation before reaching the fairness bound. The main mathematical model does not normalize fatigue by skill flexibility since greater skill flexibility does not, by itself, indicate that bearing more fatigue is equitable. The threshold at which Constraint (28) first excludes all optimal solutions of the baseline model can be characterized exactly. As this finding mainly establishes a feasibility preservation threshold rather than a new behavioral mechanism, its formal statement and proof are provided in Appendix **??**.

3.4 Fatigue-induced quality degradation

Next, we incorporate quality degradation into the model as a consequence of fatigue exposure. Therefore, the causal pathway is

$$U_{wd} \longrightarrow F_{wd} \longrightarrow \text{quality degradation.}$$

Let $R_{wisd}^{\mathrm{m}} \geq 0$ and $R_{wisd}^{\mathrm{h}} \geq 0$ denote the portions of worker w's workload assignment to task i, skill s, on day d performed while the worker is in the moderate and high fatigue states, respectively. As $\alpha_{wd} - \beta_{wd}$ equals one only in the moderate fatigue regime, the following constraints provide an exact linearisation:

$$R^{\mathrm{m}}_{wisd} \le Y_{wisd} \qquad \forall w,i,s,d \tag{29}$$

$$R^{\mathrm{m}}_{wisd} \le h_{wd}(\alpha_{wd} - \beta_{wd}) \qquad \forall w,i,s,d \tag{30}$$

$$R^{\mathrm{m}}_{wisd} \ge Y_{wisd} - h_{wd}(1 - \alpha_{wd} + \beta_{wd}) \qquad \forall w,i,s,d \tag{31}$$

$$R^{\mathrm{h}}_{wisd} \le Y_{wisd} \qquad \forall w,i,s,d \tag{32}$$

$$R^{\mathrm{h}}_{wisd} \le h_{wd}\beta_{wd} \qquad \forall w,i,s,d \tag{33}$$

$$R^{\mathrm{h}}_{wisd} \ge Y_{wisd} - h_{wd}(1 - \beta_{wd}) \qquad \forall w,i,s,d \tag{34}$$

$$R^{\mathrm{m}}_{wisd}, R^{\mathrm{h}}_{wisd} \ge 0 \qquad \forall w,i,s,d \tag{35}$$

As a result, once worker w is in the low-fatigue state, both fatigue assignment variables are zero. Under moderate fatigue, $R^{\mathrm{m}}_{wisd} = Y_{wisd}$ and $R^{\mathrm{h}}_{wisd} = 0$, while under high fatigue, $R^{\mathrm{h}}_{wisd} = Y_{wisd}$ and $R^{\mathrm{m}}_{wisd} = 0$.

Let $Q_{id} \ge 0$ indicate expected quality shortfall, expressed in equivalent rework units, associated with task i on day d. We define

$$Q_{id} = \mathrm{se}_i \sum_{w\in\mathscr{W}} \sum_{s\in\mathscr{S}} \left(v^{\mathrm{m}} R^{\mathrm{m}}_{wisd} + v^{\mathrm{h}} R^{\mathrm{h}}_{wisd}\right) \qquad \forall i \in \mathscr{I}, d \in \mathscr{D} \tag{36}$$

where se_i captures the sensitivity of task$i$ to fatigue-related execution errors and $v^{\mathrm{h}} > v^{\mathrm{m}} \ge 0$ indicate quality-loss rates under moderate and high fatigue.

The three-level specification is a tractable approximation to a general monotone fatigue–quality response. The underlying fatigue state remains continuous; only its result on quality degradation is represented piecewise.

When activity i is not in progress, Constraint (10) implies $Y_{wisd} = 0$ for each worker w and skill s, so $Q_{id} = 0$. Furthermore, subcontracted workload is not allocated to an internal labor and therefore does not inherit the internal worker fatigue state.

To internalize the economic consequence of quality shortfall, Constraint (7) is replaced by

$$\sum_{d\in\mathscr{D}} \sum_{i\in\mathscr{I}} \sum_{s\in\mathscr{S}} \left(c^{\mathrm{int}}_s \sum_{w\in\mathscr{W}} Y_{wisd} + c^{\mathrm{sub}}_s O_{isd} \right) + C^{\mathrm{del}} T + c^{\mathrm{rw}} \sum_{i\in\mathscr{I}} \sum_{d\in\mathscr{D}} Q_{id} \le b \tag{37}$$

where c^{rw} indicates the cost per unit of expected rework.

Thus, the quality degradation mechanism generates an endogenous economic cost of assigning work to fatigued internal staff. This cost interacts with workload redistribution, recovery, and subcontracting in the integrated mathematical model. A supporting finding characterizing the critical rework cost coefficient above which all optimal solutions are excluded by the quality-adjusted budget is provided in Appendix **??**.

3.5 Fatigue-induced absenteeism

The last human-centric mechanism captures the impact of fatigue on future worker availability. The timing of this relationship is crucial. The assigned workload on day d cannot cause the worker to have been absent from that same

assignment. Instead, the assigned workload contributes to the fatigue state on day d, and this state influences the risk of absence in subsequent periods. Hence, the resulting mechanism introduces a lagged feedback from current decisions to future available worker capacity.

Let π_w, π_w^{mod}, and π_w^{high} indicate worker w's calibrated absence probabilities following low, moderate, and high fatigue states, respectively, satisfying

$$0 \leq \pi_w \leq \pi_w^{\text{mod}} \leq \pi_w^{\text{high}} < 1. \tag{38}$$

To initialize incorporate the absenteeism risk, let α_{w0} and β_{w0} be fixed parameters denoting the fatigue state associated with the pre-project fatigue state F_w^0. Hence, the day-one absence probability is

$$p_{w1} = \pi_w + \left(\pi_w^{\text{mod}} - \pi_w\right) \alpha_{w0} + \left(\pi_w^{\text{high}} - \pi_w^{\text{mod}}\right) \beta_{w0} \qquad \forall w \in \mathscr{W} \tag{39}$$

If workers are assumed to enter the planning horizon in the low-fatigue state, then F_w^0 lies in the low-fatigue regime, $\alpha_{w0} = \beta_{w0} = 0$, and $p_{w1} = \pi_w$.

For subsequent project days, the risk of absence is determined by the fatigue state inherited from the preceding day:

$$p_{w,d+1} = \pi_w + \left(\pi_w^{\text{mod}} - \pi_w\right) \alpha_{wd} + \left(\pi_w^{\text{high}} - \pi_w^{\text{mod}}\right) \beta_{wd} \qquad \forall w \in \mathscr{W},\ d = 1, \ldots, D-1 \tag{40}$$

Therefore, equation (40) implies

$$F_{wd} \longrightarrow p_{w,d+1},$$

rather than a contemporaneous relationship of the form

$$U_{wd} \longrightarrow p_{wd}.$$

As fatigue itself is persistent through Equation (17), assigned workload on day d may affect absenteeism over several subsequent periods even though the direct absence mapping operates with a one-period lag.

For tactical project planning, the absenteeism probability is translated into risk-adjusted worker capacity throughout:

$$H_{wd}^{\text{eff}} = h_{wd}\,(1 - p_{wd}) \qquad \forall w \in \mathscr{W},\ d \in \mathscr{D} \tag{41}$$

and workload assignments are required to respect

$$U_{wd} \leq H_{wd}^{\text{eff}} \qquad \forall w \in \mathscr{W},\ d \in \mathscr{D} \tag{42}$$

The quantity H_{wd}^{eff} is an anticipated risk-adjusted planning capacity rather than a claim that an individual labor supplies a fractional number of hours when an absence realization occurs. Therefore, the formulation represents a deterministic expected-capacity approximation relative to tactical project planning. Explicit stochastic realization of absenteeism and recourse following an absence are outside the scope of the present paper and are suggested as a future research direction to develop this framework.

The regime transitions have a direct operational interpretation. If worker w moves from the low to the moderate fatigue state on day d, the risk-adjusted capacity available on day $d+1$ decreases by

$$\Delta H_{w,d+1}^{L\to M} = h_{w,d+1}\left(\pi_w^{\text{mod}} - \pi_w\right) \tag{43}$$

Similarly, moving from the moderate to the high fatigue state produces the additional capacity reduction

$$\Delta H_{w,d+1}^{M\to H} = h_{w,d+1}\left(\pi_w^{\text{high}} - \pi_w^{\text{mod}}\right) \tag{44}$$

Therefore, crossing a fatigue threshold generates a discrete loss of future capacity even though fatigue produced on day d does not reduce the capacity available for the workload already assigned on that day. When the absenteeism mechanism is active, Constraint (11) becomes redundant since $H_{wd}^{\text{eff}} = h_{wd}(1-p_{wd}) \le h_{wd}$. Thus, the resulting feedback structure is as $U_{wd} \longrightarrow F_{wd} \longrightarrow p_{w,d+1} \longrightarrow H_{w,d+1}^{\text{eff}}$.

The lagged formulation removes the contemporaneous feedback between today's workload and today's available capacity. Alongside this, as fatigue continues across periods, workload performed today can reduce working capacity over several future days until sufficient recovery occurs. Therefore, fatigue-induced absenteeism generates an intertemporal capacity outcome that can lead to workload redistribution, recovery, or external sourcing valuable before nominal internal capacity is exhausted.

3.6 Integrated human-centric formulation

Combining the preceding components gives the following integrated structure:

$$U_{wd} \longrightarrow E_{wd} \longrightarrow F_{wd} \longrightarrow \begin{cases} B_w^F/C_w & \text{fairness} \\ Q_{id} & \text{quality degradation} \\ p_{w,d+1} \longrightarrow H_{w,d+1}^{\text{eff}} & \text{future workforce capacity} \end{cases}$$

Therefore, the individual human-centric structures are connected through a fatigue state. This structure allows comparative statics across fairness, quality, and absenteeism. In particular, the persistence parameter ρ simultaneously governs the extent to which today's workload moves forward into all three downstream consequences.

The first integrated finding characterizes the impact of slower fatigue recovery on the feasibility of the model.

PROPOSITION 4 **(Monotonicity in fatigue persistence)**. *Consider two persistence parameters respecting*

$$0 \le \rho_1 \le \rho_2 < 1,$$

with all other parameters and the initial fatigue states remain unchanged. For any fixed internal assignment trajectory $\{Y_{wisd}\}_{w,i,s,d}$*, and hence its induced workload trajectory,*

$$F_{wd}(\rho_1) \le F_{wd}(\rho_2) \qquad \forall\, w \in \mathscr{W}, d \in \mathscr{D}$$

Under the monotone fatigue mappings defined in the previous subsections, cumulative fatigue pressure and fatigue-induced quality shortfall are therefore weakly larger under ρ_2*, whereas risk-adjusted future labor capacity is weakly smaller.*

Let $\mathscr{X}^{HC}(\rho)$ *indicate the feasible set of the integrated human-centric model projected onto the original staffing and scheduling variables. Then*

$$\mathscr{X}^{HC}(\rho_2) \subseteq \mathscr{X}^{HC}(\rho_1),$$

and consequently

$$Z^{HC}(\rho_1) \le Z^{HC}(\rho_2).$$

Proposition 4 demonstrates a direct connection between worker recovery characteristics and project performance. Slower recovery cannot enlarge the feasible region and therefore cannot improve the optimal objective. Importantly, these effects do not necessarily translate immediately into project delay. For moderate levels of persistence, the model may maintain the project completion time through redistribution of workload, additional recoveries, or more subcontracting. Only when these adjustments are insufficient does greater persistence necessarily translate into a longer project completion time.

The integrated mathematical model also changes the local comparison between using internal staff and subcontracted labor. In the baseline model, replacing subcontracted work with internal labors can reduce direct staffing cost when $c_s^{\text{int}} < c_s^{\text{sub}}$. However, in the integrated human-centric model, assigning additional work to an already fatigued internal worker can generate rework cost, escalate cumulative fatigue pressure, and increase absenteeism risk. The following result provides a conservative sufficient condition under which substituting subcontracted worker for fatigued internal work is noninferior even when nominal internal capacity remains available.

PROPOSITION 5 **(Dominance condition for subcontracting fatigued work)**. *Consider a feasible solution in which worker w performs* $\delta > 0$ *units of task i, requiring skill s, on day d while is under fatigue state* $r \in \{\text{mod}, \text{high}\}$*. Suppose that these* δ *units can be replaced by subcontracted worker without changing activity duration, required skill, or any other exogenous sourcing requirement.*

Define the subcontracting premium as $\Delta c_s = c_s^{\text{sub}} - c_s^{\text{int}}$*, and let*

$$v^r = \begin{cases} v^{\text{m}} & r = \text{mod} \\ v^{\text{h}} & r = \text{high} \end{cases}$$

If $\Delta c_s \le c^{\text{rw}} \text{se}_i v^r$*, then replacing the* δ *units of fatigued internal work with subcontracted worker generates another feasible solution with the same project completion time and:*

1. *no greater total budget consumption;*
2. *a strictly smaller cumulative fatigue pressure for worker w; and*

3. *weakly lower subsequent absenteeism risk, and consequently, weakly higher future effective working capacity for worker w.*

The condition in Proposition 5 is conservative since it compares the subcontracting premium only with the immediate rework cost avoided by eliminating the fatigued internal labor assignment. It does not allocate a monetary value to the accompanying reduction in cumulative fatigue, the relaxation of the fairness cap, or the raise in future internal capacity. Furthermore, if the workload reduction changes worker w from a higher fatigue state into a lower fatigue regime, quality shortfalls associated with the worker's remaining assignments may also decrease. As a result, a subcontracting substitution may stay operationally noninferior even when $\Delta c_s > c^{\mathrm{rw}} \mathrm{se}_i v^r$. Thus, Proposition 5 should not be interpreted as stating that the model must subcontract whenever the sufficient condition holds. Instead, this proposition establishes that available internal capacity does not by itself imply that using that capacity is operationally preferable: under the stated condition, there exists a solution with the same objective, no greater budget consumption, and strictly lower fatigue exposure obtained by substituting subcontracted worker.

The structural findings developed across the model clarify the different channels through which fatigue influence scheduling and staffing decisions. Lemma 1 establishes that current workload has persistent impacts beyond the period in which it is assigned. Proposition 1 quantifies the required recovery to leave an elevated fatigue state, while Proposition 2 demonstrates that temporally flexible work should be organized around the resulting fatigue trajectory instead of equal working durations alone. Proposition 3 further indicates that workload timing influences cumulative fatigue pressure. In the integrated human centric model, Proposition 4 establishes how slower recovery progressively constraints feasible project operations, whereas Proposition 5 identifies a condition under which fatigue externalities can make subcontracting noninferior before exhausting internal capacity.

Collectively, these findings indicate that fatigue is not simply an additional workforce restriction. It is an endogenous intertemporal state that reshape the operational value of workload timing, recovery, workforce allocation, and sourcing decisions. All proofs are provided in the Online Appendix (e-companion). Now, Figure 2 is provided to summarize the temporal mechanism underlying the integrated human-centric formulation. The figure illustrates how workload assignment decisions interact with fatigue and recovery across consecutive project days, and how the resulting fatigue state influences quality degradation, fairness, and future availability of workers.

As shown in Figure 2, the fatigue state on day d is determined jointly by residual fatigue carried from the previous day and the new workload exposure produced by the workload allocation decision. Recovery reduces the portion of the previous fatigue that persists into the current period, while new allocation of workload produces additional fatigue burden. The resulting fatigue state then influences three distinct outcomes: fatigue-induced quality degradation, fairness in distribution of cumulative fatigue burden, and the absenteeism risk on the following day. The absenteeism risk creates the intertemporal feedback loop in the proposed framework. The fatigue state on day d determines the probability of absence for day $d+1$, which in turn determines the effective working capacity available on that day. This capacity restricts the workload that can be assigned to the labor on day $d+1$. The new workload exposure, jointly with the residual portion of the fatigue state carried forward from day d, then determines the fatigue on day $d+1$. Therefore, decisions on the current workload allocation can affect both current human-centric outcomes and future staffing flexibility through the persistence of fatigue state.

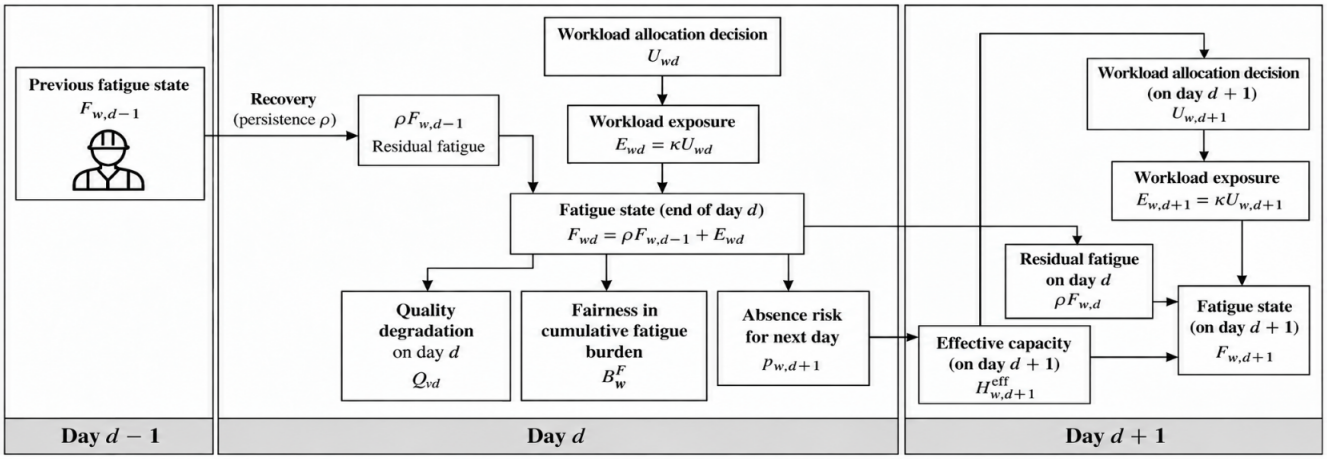


Figure 2 Dynamic mechanism connecting workload allocation, fatigue, human-centric consequences, and future workforce capacity.

4 Experiment results

The proposed model is evaluated using a residential construction case study adapted from Hosseinian and Baradaran (2023), which provides a realistic representation of project operations. The construction activities are linked through finish-to-start precedence relations, and the project structure incorporates multiple skill categories to capture the heterogeneous skill requirements typical of residential construction projects. The core project data, including task definitions, precedence links, and required skills, are derived from the referenced case study. The first part of the results is related to incorporating fatigue monitoring mechanism into basic model. Although fatigue is not yet internalized in decision-making at this stage, the resulting patterns provide important managerial insights. Table 1 shows that fatigue exposure is highly concentrated among a subset of workers. A small group of heavily utilized workers performs a large proportion of their assignments under fatigue conditions, while others experience little to no fatigue despite participating in the schedule. The decomposition further reveals that fatigue exposure is mechanism-dependent. For some workers, fatigue is primarily driven by intensive daily workloads, whereas for others it arises from insufficient recovery across consecutive working days. This heterogeneity highlights that fatigue is not only unevenly distributed across the workforce, but also driven by distinct operational factors. These findings indicate that, when only project outcomes are considered and fatigue consequences are not incorporated, staffing decisions may lead to concentrated and uneven fatigue exposure. This motivates the subsequent model extensions, which explicitly incorporate the impacts of fatigue and introduce mechanisms to balance workload distribution and mitigate excessive fatigue accumulation.

We next incorporate the fairness structure to examine how bounding fatigue disparities reshapes workload allocation across workers. Figure 3 compares worker-level fatigue exposure, measured by (**??**), with and without fairness constraints. The results show a substantial compression in the dispersion of fatigue exposure. Without fairness, fatigue is highly concentrated among a small subset of workers, particularly those whose skills are critical for project execution. These workers experience significantly higher fatigue levels relative to the rest of the workforce, indicating a concentration of workload and elevated risk of burnout. In contrast, when fairness constraints are enforced, fatigue exposure is redistributed more evenly across workers, leading to a markedly narrower range

Table 1 Worker-level fatigue exposure and root-cause decomposition

Worker	Worked (days)	Worked under fatigue (%)	Fatigue– Consecutive days (%)	Fatigues– Daily workload (%)
W6	23	95.7	45.2	54.8
W5	22	86.4	48.5	51.5
W7	14	78.6	0.0	100
W8	10	90.0	12.5	87.5
W1	9	66.7	40.0	60.0
W2	13	46.2	44.4	55.6
W3	12	58.3	33.3	66.7
W4	19	73.7	35.7	64.3
W10	3	66.7	0.0	100
W9	3	0.0	–	–

of fatigue scores. This redistribution reflects a structural shift in staffing decisions. The model reduces reliance on a limited group of highly demanded workers and instead allocates tasks more broadly across the available workforce, even if this requires utilizing less frequently assigned workers or adjusting subcontracting decisions. As a result, workload pressure is no longer concentrated on a few individuals, but is shared more equitably. Importantly, this rebalancing is achieved while preserving feasibility and maintaining comparable project delivery performance. From a managerial perspective, the results demonstrate that fairness constraints act as a mechanism for mitigating concentrated fatigue risk by smoothing workload allocation, particularly in the presence of skill bottlenecks. In the baseline setting, these bottlenecks lead to repeated utilization of specific workers; the fairness structure explicitly counteracts this tendency by limiting allowable disparities in fatigue exposure. Overall, incorporating fairness transforms staffing decisions from a concentrated allocation pattern toward a more balanced workload distribution, reducing extreme fatigue exposure without compromising the ability to deliver the project. This indicates that fairness constraints effectively relax implicit skill bottlenecks by forcing diversification in task assignments.



Figure 3 Comparing the spread of fatigue exposure across workers with and without a fairness structure. Orange and blue bars show the fairness score of each worker without and with fairness, respectively.

The managerial implications and operational decisions related to the comparison of incorporating a fairness structure in the model against the basic model are illustrated in Figure 4. The blue frontier, $\Gamma(C)$, shows the

minimum required budget for project completion time of C without considering fairness, while the red frontier, $\Gamma^{\text{fair}}(C)$, indicates the corresponding budget after incorporating a fairness mechanism. The intersections of these frontiers with the fixed budget b determine the optimal project completion time with and without considering fairness, represented by $C_*^{\text{fair}}(b)$ and $C_*^0(b)$, respectively. The horizontal gap between these two points demonstrates the project delay caused by incorporating fairness, while the vertical gap between the frontiers shows the additional budget consumption due to incorporating fairness. This figure indicates how fairness constraints reshape the budget–completion-time frontier and modify the feasible schedules.

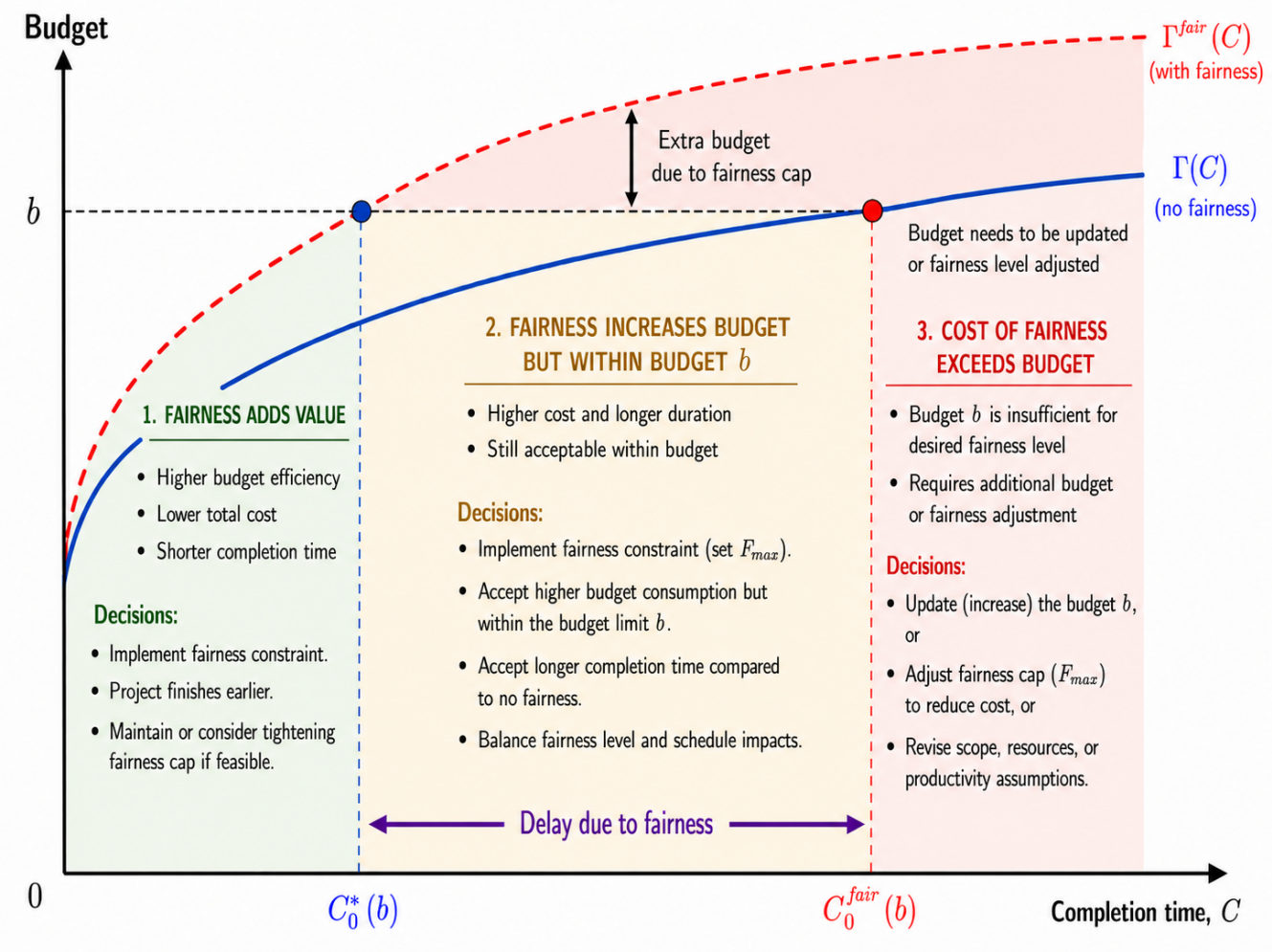


Figure 4 Managerial implications and operational decisions of incorporating fairness.

Figure 4 identifies three regions with distinct managerial implications and operational decisions. In the first region, fairness can be incorporated without threatening schedule feasibility or exceeding the available budget. This situation typically arises in construction projects, and workload redistribution among workers can be achieved with limited operational disruption. In the second region, incorporation of fairness maintains operational feasibility, but only at the expense of longer project delivery. In practice, this happens when, due to a shortage of workers, specialized skill requirements, and project network, the model is restricted to balance workloads efficiently, forcing construction project managers to rely on extending project deadlines, subcontracting, or additional staffing costs

to maintain fairness targets. In the third region, the fairness-adjusted budget frontier exceeds the available budget, revealing that the desired fairness limit is no longer financially feasible under the current project configuration. In such conditions, decisions must be revisiting the fairness cap, allocating additional budgets, extending deadlines, or redesigning staffing and subcontracting strategies. From an operations management perspective, figure 4 indicates that fairness incorporation directly reshapes the budget–time trade-off structure of construction projects rather than acting as a purely behavioral or human-resource consideration. The magnitude of the fairness-induced frontier shift depends not only on the fairness cap but also on project operational characteristics such as precedence relations, resource intensity, skill specialization, and worker flexibility. Consequently, this figure provides project managers with a practical decision-support framework to evaluate when fairness can be operationally absorbed within the existing project structure and when additional budget or scheduling interventions become necessary.

The next part of the results is associated with the incorporation of absenteeism to examine how excess workload translates into endogenous reductions in effective workforce capacity and how this affects quality degradation. Figure 5 compares daily quality degradation with and without incorporating absenteeism. The results show that incorporating absenteeism leads to a clear reduction in both the magnitude and variability of quality loss over time. When absenteeism is not modeled, workload concentration results in pronounced spikes in quality degradation. In contrast, once absenteeism is incorporated, the model anticipates potential capacity losses and adjusts workload allocation accordingly. This results in fewer instances of excessive workload and, consequently, fewer periods in which workers operate under fatigue conditions. Importantly, the improvement in quality is not imposed through additional constraints on quality itself, but emerges endogenously. By internalizing absenteeism risk, the model avoids assigning workloads that would elevate fatigue to levels associated with higher absence probability. This anticipatory adjustment reduces both fatigue exposure and the associated quality degradation. From a managerial perspective, these findings highlight that incorporating absenteeism acts as a proactive stabilization mechanism. It not only improves schedule reliability by accounting for workforce uncertainty, but also enhances quality outcomes by mitigating excessive workload concentration. Overall, modeling absenteeism transforms staffing decisions by linking workload allocation to future capacity risk, leading to more balanced schedules with lower fatigue exposure and improved quality performance.

Figure 6 is presented to show the endogenous capacity mechanism generated by incorporating absenteeism. Unlike classical operations management and construction scheduling approaches, where the capacity of workers to perform jobs is typically treated as a fixed exogenous input, the developed framework models effective workforce availability as a function of absenteeism risk. In this regard, the available effective working duration of workers depends on the interaction between allocated workload and the resulting absenteeism probability. As a result, workers' capacity becomes endogenously determined in the scheduling process instead of being assumed to be constant throughout project execution. The red curve in Figure 6 indicates the effective working duration under absenteeism, $H_{wd}^{\text{eff}} = h_{wd}(1 - p_{wd})$. The blue dashed line is the identity line, $y = U_{wd}$, which shows the point at which the effective working duration equals the assigned workload. The intersection point between these two defines the endogenous upper bound U_{wd}^{*}, characterized by $U_{wd}^{*} = h_{wd}\big(1 - p_{wd}(U_{wd}^{*})\big)$.

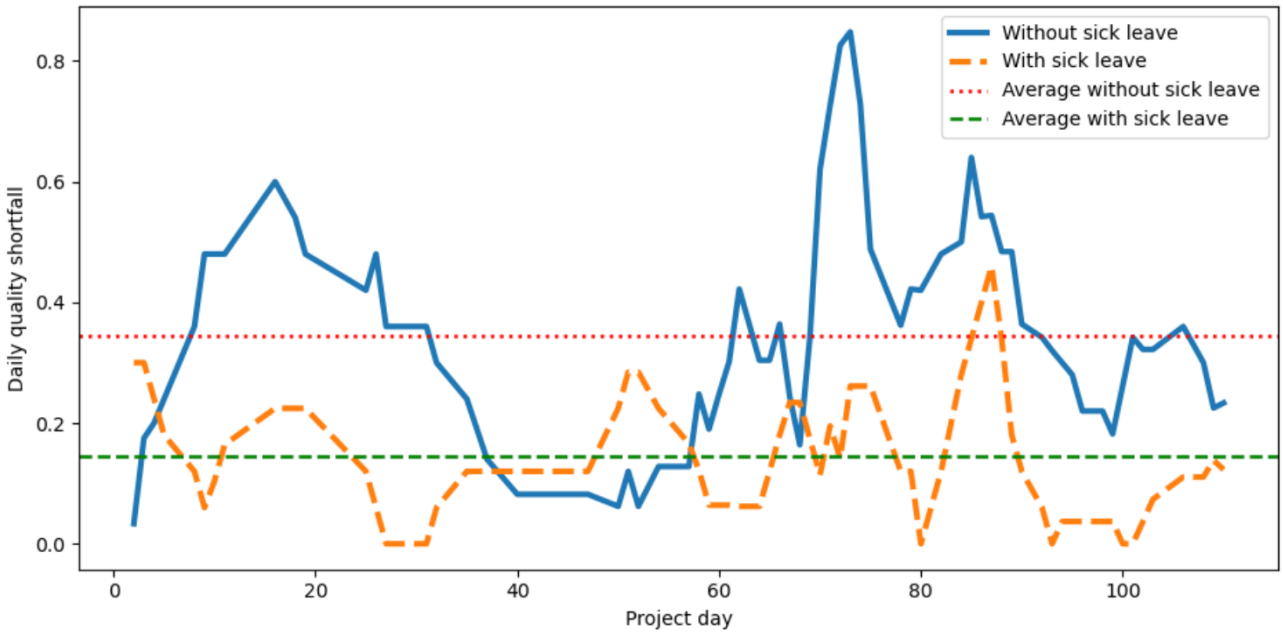


Figure 5 The comparison of daily quality loss in the baseline mathematical model and fatigue-oriented sick leave model reduced effective capacity. Horizontal lines show the average quality degradation in each case.

This fixed point indicates the maximum workload that can be allocated to workers while remaining operationally sustainable after incorporating absenteeism impacts. Accordingly, the feasible region corresponds to the allocated workload that satisfies. $U_{wd} \leq H_{wd}^{\text{eff}}$, where effective working duration remains enough to meet the assigned workload. In contrast, the cannot be maintained whenever $U_{wd} > H_{wd}^{\text{eff}}$, showing that absenteeism capacity erosion reduces the capacity of available workers below the allocated workload level.

From an operations management standpoint, this figure indicates that staff capacity should not be interpreted as a static parameter when heavy workload influences the risk of absenteeism among the workforce. In construction projects, excessive workload assignments may increase fatigue of workers and consequently increase the risk of higher absenteeism, which subsequently reduces effective staff capacity and reshapes the attainable scheduling region. Therefore, workload allocation beyond U_{wd}^{*} creates unstable schedules that cannot be reliably sustained in practice, even if they appear feasible under classical planning assumptions. Furthermore, Figure 6 also demonstrates the capacity loss ignored when sick-leave risk is not explicitly formulated in staffing and scheduling decisions. The vertical gap between the nominal capacity level h_{wd} and the absenteeism-adjusted effective working duration H_{wd}^{eff} indicates the portion of workforce capacity that becomes unavailable due to a higher risk of fatigue-induced absenteeism. Traditional construction scheduling procedures typically neglect this hidden reduction in staff working capacity and therefore may systematically overestimate attainable workforce capacity. By explicitly formulating the risk of absenteeism into the staffing and scheduling decisions, the proposed model generates more practically attainable schedules that better reflect the dynamic interaction between workload allocation, human factors, and effective workforce performance. To synthesize the results, Table 2 compares system-level outcomes across progressively richer model variants that incorporate human-centric mechanisms. The scenarios range from a baseline setting that only monitors fatigue (Scenario 1) to models that sequentially incorporate fairness constraints (Scenario 2), quality degradation (Scenario 3), and endogenous absenteeism (Scenario 4).

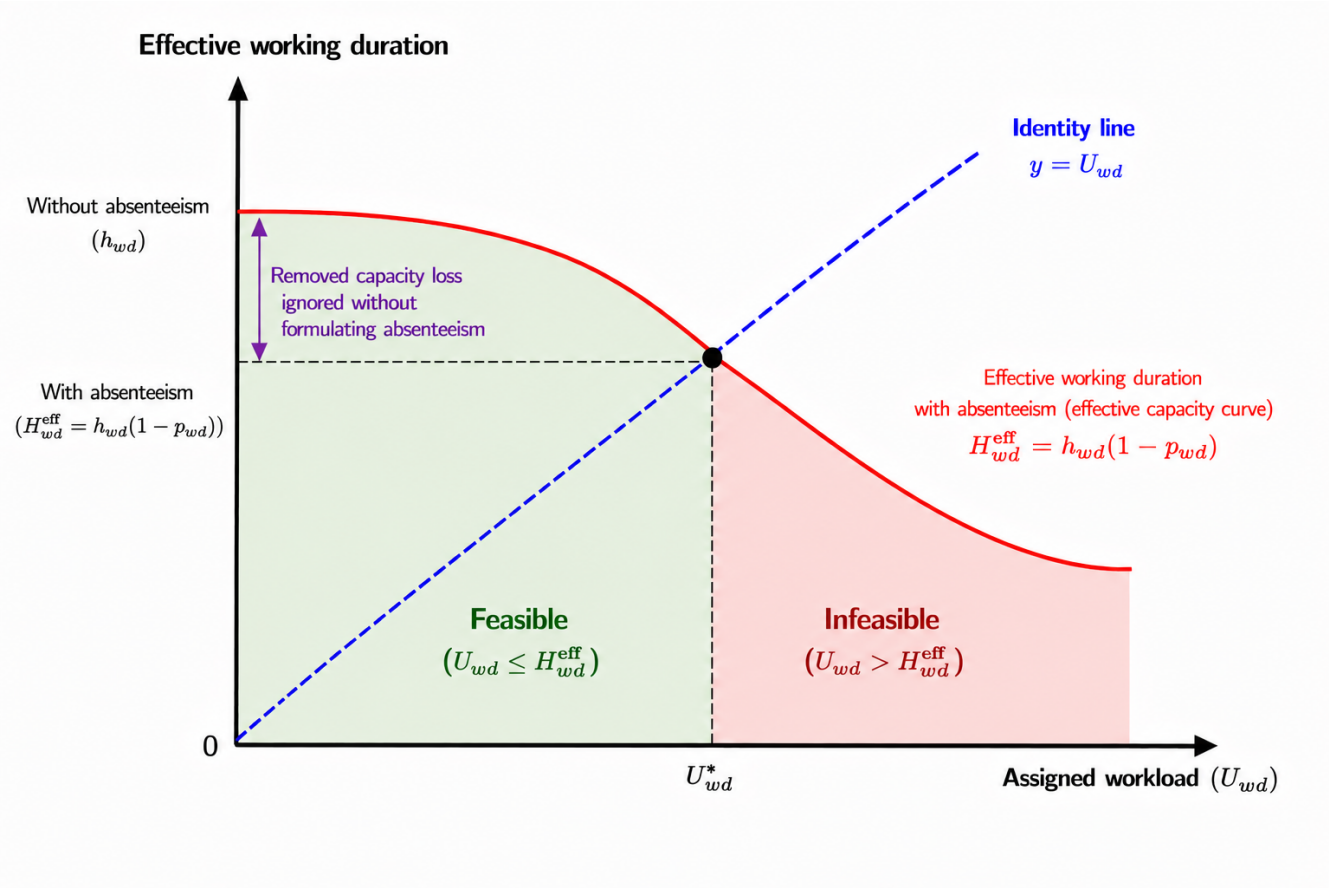


Figure 6 Endogenous effective workforce capacity by incorporating absenteeism and the resulting feasible and infeasible regions.

Table 2 Comparison of the results under various extensions

Metric	Scenario 1	Scenario 2	Scenario 3	Scenario 4
Total productive hours	1692.0	1695.1	1731.3	1691.4
Regular hours share (%)	45.4	43.1	52.6	42.4
Subcontractor hours share (%)	54.6	56.9	47.4	57.6
Worked days	128	185	232	193
Worked days with moderate fatigue (%)	75.0	36.8	50.4	27.5
Worked days with high fatigue (%)	45.3	3.2	12.1	1.6
Fatigue due to workload (%)	86.5	73.5	65.8	67.9
Fatigue due to consecutive days (%)	46.9	39.7	63.2	58.5
$\min_w F_w$	5	7	16	5
$\max_w F_w$	23	9	18	7
ΔF (max–min)	18	2	2	2
Total quality shortfall	30.6	29.4	24.1	10.2
Peak daily quality shortfall	1.8 (day 71)	1.4 (day 86)	1.4 (day 72)	0.8 (day 86)

The results reveal a fundamental distinction between nominal project performance and sustainable operational outcomes. When only project outcomes are considered (Scenario 1), the model produces highly compressed schedules with lower apparent effort requirements. However, this comes at the cost of widespread fatigue exposure and concentrated workload allocation. Incorporating fairness and quality considerations (Scenarios 2 and 3) redis-

tributes workload across workers and extends the project duration, reflecting a shift from aggressive scheduling toward more sustainable execution. Interestingly, incorporating absenteeism (Scenario 4) does not increase total effort, but instead moderates workload allocation by reducing effective capacity. This leads to more conservative and realistic schedules without inflating total labor usage. In this sense, absenteeism acts as an endogenous regulator that prevents over-utilization rather than simply adding additional workload. The results show that subcontracting is not merely a cost-driven decision, but an endogenous mechanism for managing operational risks. Its utilization increases when internal workforce constraints become binding due to fairness limits or reduced effective capacity. In contrast, when quality degradation is explicitly modeled (Scenario 3), the model shifts toward greater use of internal workers by spreading workload over a longer horizon. This highlights a practical trade-off: subcontracting supports short-term flexibility and reliability, while extended scheduling horizons enable safer utilization of internal resources. Fatigue patterns differ significantly across scenarios. Under the baseline setting, both moderate and high fatigue are prevalent, indicating systematic overload when fatigue consequences are not internalized. Introducing human-centric mechanisms substantially reduces high fatigue exposure, with the most pronounced reduction observed when absenteeism is incorporated. This reflects the model's tendency to avoid workload allocations that would trigger elevated fatigue and associated risks.

At the same time, moderate fatigue remains present across all scenarios, reflecting the inherent labor intensity of construction activities. Decomposition results indicate that daily workload intensity is the primary driver of fatigue, while fatigue due to consecutive working days becomes more pronounced in longer project schedules. This suggests that both workload smoothing and recovery planning are essential for managing fatigue effectively. The incorporation of human-centric mechanisms leads to significant improvements in both fairness and quality outcomes. Without fairness constraints, fatigue exposure is highly uneven across workers. Once fairness is introduced, the dispersion of fatigue exposure is sharply reduced, preventing concentration of workload on a small subset of workers. Quality outcomes exhibit a similar pattern. Total and peak quality losses are highest in the baseline model and decrease progressively as additional mechanisms are incorporated. Notably, the absenteeism-aware model achieves the lowest quality shortfall, despite not directly constraining quality. This improvement arises from anticipatory workload adjustments that reduce fatigue exposure and prevent extreme workload concentrations.

Figure 7 is presented to summarize the decision logic of the developed framework and indicates how each human-centric layer shifts the optimal solution of the basic integrated staffing and scheduling model. This figure focuses on two managerial outcomes that are directly related to project planners: project completion time and reliance on subcontractors. In this figure, T_0 and R_0 show the project completion time and subcontracting reliance of the basic model, respectively. Here, T_1 and R_1 denote the project delivery time and subcontracting reliance under the fairness-regulated model, T_2 and R_2 indicate the corresponding outcomes under the quality loss model, and T_3 and R_3 show the results under the absenteeism model.

Figure 7 provides a practical decision-support framework for interpreting the results of the developed framework. The first layer demonstrates that fatigue monitoring alone does not change the optimal project completion time and subcontracting reliance when it is used only as an observation mechanism. Hence, measuring fatigue without

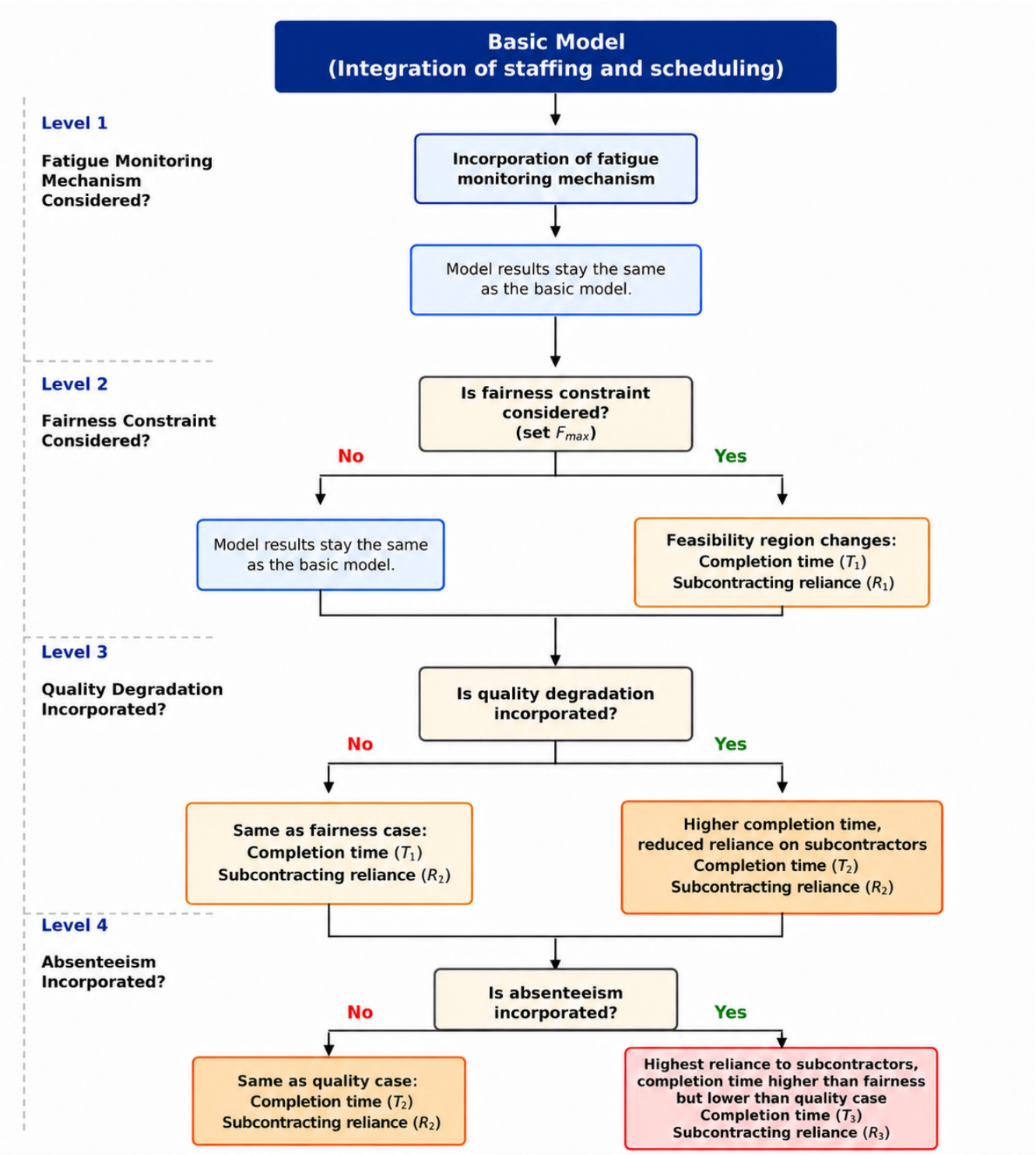


Figure 7 Decision flow of incorporating fatigue monitoring, fairness limit, quality degradation, and absenteeism into the basic model.

incorporating it into constraints or the objective function leaves the solution equivalent to the basic model. The first structural change in the results occurs when fairness is formulated through the fairness cap $F^{\max}$. Here, some feasible schedules in the basic model may no longer be acceptable as they impose uneven fatigue burdens across workers. Therefore, the feasible region contracts, project delivery time increases from T_0 to T_1, and subcontracting reliance shows an increasing trend moving from R_0 to R_1 because it is required to use more external labor to

redistribute workload and preserve fairness within internal workers. The next layer incorporates quality loss caused by working under fatigue. This layer reshapes the economic logic of the model as overloaded internal workers may generate rework or quality degradation costs. Hence, the model avoids some workload-intensive staffing decisions that were previously attractive under the fairness-only regime. As a result, it leads to the longest project delivery time, T_2, since the project schedule absorbs rework penalties by slowing down or redistributing workload. However, subcontracting reliance decreases to R_2, because the model internalizes the cost of quality degradation and may choose less aggressive subcontracting or workload concentration when such decisions increase downstream rework costs. The final layer formulates absenteeism to the model, which links effective worker capacity endogenously to fatigue exposure. Unlike the quality-oriented model, absenteeism reduces expected available capacity, which leads to a higher need for workers. The absenteeism model shows the highest reliance on subcontractors, R_3, since subcontractors become a primary operational lever for compensating for fatigue-based capacity erosion. However, the project delivery time under absenteeism, T_3, is lower than the completion time of the quality-loss model T_2, although still higher than the fairness and basic models. In summary, the directional ordering is $T_2 > T_3 > T_1 > T_0$, $R_3 > R_1 > R_0 > R_2$. This ordering reveals a central operational insight of the study: different human-centric factors influence the project scheduling and staffing decisions through different channels. Fairness mainly restricts the feasible set, quality degradation increases the cost of intensive schedules, and absenteeism erodes effective capacity. Thus, the managerial response cannot be reduced to a single "more human-centric means more costly or longer project duration" conclusion. Instead, each human-centric factor changes the optimal balance between schedule duration, internal staff utilization, and subcontractor reliance in a distinct way.

5 Conclusion

The effect of workload, fatigue, and fairness on operational performance has attracted increasing attention in human-centric operation management. However, existing models in construction and operation management often treat workforce capacity as an exogenous input and rarely analyze how workload allocation influences fatigue exposure, quality outcomes, and sick leave in construction projects. This paper contributes to this growing body of literature by introducing an integrated staffing and scheduling model for residential construction projects in which fatigue exposure of workers, fairness in fatigue exposure, quality degradation, and fatigue-induced absenteeism are embedded directly into operational decision-making. We show that human-centric factors influence staffing and scheduling decisions through distinct operational channels. The fatigue-monitoring mechanism tracks worker fatigue exposure without directly altering the optimal solution structure. The fairness extension restricts excessive imbalance in experiencing fatigue across workers and therefore contracts the feasible region. The quality degradation layer converts excess workload into endogenous quality shortfall and rework cost, shifting the time-cost frontier upward. Most importantly, the absenteeism risk layer links fatigue exposure to the effective working capacity of staff, such that increasing workload elevates sick leave risk and simultaneously reduces expected productive capacity. This feedback mechanism reshapes the structure of staffing decisions by making workforce capacity endogenous to the assigned workload.

Our computational results based on a residential case study provide several managerial and operational insights. First, if human-centric factors are ignored, the model tends to concentrate workload on a limited subset of workers with critical skills, resulting in a higher fatigue experience, uneven workload allocation, and increased risk of quality degradation and absenteeism. Second, the introduced model is formulated in a way that identifies not only which workers and skills should be used internally, but also when subcontracting is economically justified. In some cases, subcontracting is selected even for skills that are available in-house because it can relieve internal resource bottlenecks, reduce excessive workload pressure on workers, and improve schedule feasibility. Third, the findings demonstrate that most fatigue exposures are driven by high daily workload rather than consecutive working days, suggesting that fatigue management should not rely only on rest-day or rotation policies, but should also control daily workload assignment.

The results also highlight the importance of considering fatigue, fairness, quality, and absenteeism jointly rather than incorporating isolated extensions. Adding a fairness structure may increase project duration, but it substantially moderates allocated workloads and produces more balanced workload distribution among workers. Incorporating quality degradation alone can be misleading because the model may respond by substantially extending project completion time to reduce quality shortfall. In contrast, when fatigue-induced absenteeism is incorporated, the model can achieve shorter project duration and better quality outcomes than a quality-only model. This occurs as the model anticipates the future capacity loss associated with excessive fatigue exposure and, therefore, reallocates workload more conservatively, which in turn reduces fatigue exposure across the workforce and generates a quality-enhancing spillover effect. Thus, absenteeism risk does not merely reduce available working capacity; it creates an endogenous stabilizing mechanism that enhances workforce reliability and project performance.

From a policy perspective, the results suggest that government bodies, construction regulators, and industry associations can play a critical role in promoting fatigue-sensitive workforce planning in the construction sector. As reducing total workload by increasing workforce capacity may not always be feasible because of budget constraints, labor shortages, subcontractor availability, or delivery deadlines, regulatory and industry-level interventions may instead focus on how workload is allocated across the workforce and time periods. For example, providing guidance on maximum daily workload, fatigue-sensitive allocation rules, workload-balancing requirements, and look-ahead planning practices can help reduce excessive dependence on a small group of critical workers without needing a proportional increase in total workers. Such policies would move beyond monitoring compliance with working-hour limits and encourage construction companies to manage the distribution of workload, which our results show is central to fatigue exposure, quality degradation, and absenteeism.

From a construction firm's perspective, the findings suggest that construction firms should incorporate fatigue, fairness, quality, and absenteeism mechanisms directly into their internal rostering systems. Most fatigue activations in our experiments are driven by excessive daily workload rather than only by consecutive working days, revealing that rosters can comply with the standard working-hour limits, while still generating risk of fatigue exposure through uneven daily workload allocation. Hence, policies focused only on rest days, shift rotation, or weekly working hours may be insufficient. Companies can use fatigue-sensitive rostering systems to prevent repeated assignment of demanding tasks to the same workers and identify workers or skills exposed to persistent workload

pressure. The model also supports targeted responses such as cross-training, temporary hiring, selective subcontracting, or workload redistribution. In this sense, fatigue-sensitive rostering is not only a compliance tool but also a proactive planning mechanism to protect workforce reliability, project quality, and long-term staffing stability.

Our framework relies on several assumptions made for tractability. First, task durations, skill availability, subcontractor availability, and fatigue thresholds are treated as known inputs. We expect that fatigue experience will be associated with the type of task and environmental conditions, which are not included in our paper. Secondly, although impacts of human factors allow us to isolate the operational effects of different human-centric mechanisms, the relationships between fairness, quality degradation, and absenteeism provide opportunities for future research. Future work may build on this limitation by explicitly modeling the interactions among human-centric factors. Capturing and formulating these effects would provide a richer representation of workforce dynamics in construction operations. Another promising direction is to extend the model to multi-project or multi-site environments, where the workforce may be transferred across construction sites, and fairness concerns arise from the distribution of workload, travel burden, site assignments, and fatigue experience across projects. Future works may also introduce incentive or reward functions for workers with persistently high fatigue exposure, allowing the model to examine trade-offs among compensation, fairness, retention, absenteeism risk, and project performance.

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